

CENG381 Stochastic Processes

Queuing Theory: M/M/1 and M/M/inf — Practice 1 (Answer Key)

Q1 Solution

a) Stability and traffic intensity:

$$\rho = \lambda / \mu = 4/6 = 2/3 < 1 \Rightarrow \text{queue is stable} \quad \checkmark$$

b) Stationary distribution:

$$p_i = (1 - \rho) * \rho^i = (1/3) * (2/3)^i$$

$$p_0 = 1/3 \quad \approx 0.333 \quad (\text{no patients})$$

$$p_1 = (1/3) * (2/3) = 2/9 \quad \approx 0.222$$

$$P(N \geq 2) = 1 - p_0 - p_1 = 1 - 1/3 - 2/9 = 1 - 5/9 = 4/9 \approx 0.444$$

c) $E[N]$, $E[T]$, and Little's Law:

$$E[N] = \rho / (1 - \rho) = (2/3) / (1/3) = 2 \text{ patients}$$

$$E[T] = 1 / (\mu - \lambda) = 1 / (6 - 4) = 1/2 \text{ hour} = 30 \text{ minutes}$$

$$\text{Little's Law: } E[N] = \lambda * E[T]$$

$$\text{LHS} = 2$$

$$\text{RHS} = 4 * (1/2) = 2 \quad \checkmark$$

Q2 Solution

a) Why M/M/∞ is always stable:

In state n there are n servers working simultaneously, so the total service rate is $n\mu$ — it grows without bound as n increases. No matter how large the queue gets, the departure rate eventually exceeds the arrival rate λ , so the queue never drifts to infinity. Rate structure: birth rate $\lambda_n = \lambda$ (all n); death rate $\mu_n = n\mu$.

b) Stationary distribution derivation:

$$\text{Detailed balance: } \lambda * p_n = (n+1)\mu * p_{n+1}$$

$$\Rightarrow p_{n+1} = (\lambda / ((n+1)\mu)) * p_n = (\rho / (n+1)) * p_n$$

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Telescope: $\pi_n = \pi_0 * \rho^n / n!$

Enforce sum = 1:

$$\pi_0 * \sum_{n=0}^{\infty} \rho^n / n! = \pi_0 * e^{\rho} = 1$$

$$\Rightarrow \pi_0 = e^{-\rho}$$

$\pi_n = e^{-\rho} * \rho^n / n!$ [Poisson(ρ) distribution]

$$E[N] = \rho = \lambda / \mu$$

c) Numerical values ($\lambda=6, \mu=3, \rho=2$):

$$\rho = 6/3 = 2$$

$$\pi_0 = e^{-2} \approx 0.1353$$

$$\pi_1 = e^{-2} * 2 \approx 0.2707$$

$$\pi_2 = e^{-2} * 4/2 = 2 * e^{-2} \approx 0.2707$$

$$P(N \geq 3) = 1 - \pi_0 - \pi_1 - \pi_2$$

$$= 1 - e^{-2} * (1 + 2 + 2)$$

$$= 1 - 5 * e^{-2}$$

$$\approx 1 - 5 * 0.1353 = 1 - 0.6767 = 0.3233$$

Q3 Solution

a) Performance metrics ($\lambda=80, \mu=100$):

$$\rho = 80/100 = 0.8$$

$$\pi_0 = 1 - \rho = 0.2 \quad (20\% \text{ chance router is idle})$$

$$E[N] = \rho / (1 - \rho) = 0.8 / 0.2 = 4 \text{ packets}$$

$$E[T] = 1 / (\mu - \lambda) = 1/20 = 0.05 \text{ s} = 50 \text{ ms}$$

$$\begin{aligned} E[W] &= E[T] - 1/\mu = 1/20 - 1/100 = 5/100 - 1/100 = 4/100 \\ &= 0.04 \text{ s} = 40 \text{ ms} \end{aligned}$$

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b) Maximum arrival rate for SLA $E[W] \leq 5$ ms:

$$E[W] = \lambda / [\mu(\mu - \lambda)] \leq 0.005$$

$$\lambda / [100(100 - \lambda)] \leq 0.005$$

$$\lambda \leq 0.005 * 100 * (100 - \lambda)$$

$$\lambda \leq 0.5(100 - \lambda)$$

$$\lambda \leq 50 - 0.5\lambda$$

$$1.5\lambda \leq 50$$

$$\lambda \leq 100/3 \approx 33.3 \text{ packets/second}$$

To maintain $E[W] \leq 5$ ms, the arrival rate must not exceed $100/3 \approx 33.3$ packets/second — less than half the current rate of 80.

c) Little's Law verification and M/M/2 comparison:

$$\text{Little's Law: } E[N] = \lambda * E[T]$$

$$\text{LHS: } E[N] = 4$$

$$\text{RHS: } 80 * 0.05 = 4 \quad \checkmark$$

For M/M/2 (two servers, each at rate $\mu=100$ packets/s):

- The effective service capacity doubles to 200/s.
- With $\lambda=80$, $\rho = 80/(2 \cdot 100) = 0.4$ per server — much less loaded.
- $E[W]$ for M/M/2 is substantially smaller than for M/M/1 at $\lambda=80$, because there is a spare server to absorb bursts.

Qualitatively: adding a second server reduces congestion super-linearly — halving the load factor reduces waiting time by far more than a factor of 2 due to the nonlinear behaviour of the M/M/1 formula $E[W] = \lambda/[\mu(\mu-\lambda)]$ near saturation.